SILAS KWABLA GAH

Ph.D. Candidate in Computer Science · 3D Computer Vision & Foundation Models

Department of Computer Science, University of Ghana, Legon, Accra, Ghana

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Research Interests

Core Areas: 3D Computer Vision, Multi-View Geometry, Foundation Model Coordination ($\Delta\theta = 0$), Open-Vocabulary 3D Scene Understanding, Semantic Retention Dynamics, High-Throughput Sparse Computing, 3D Gaussian Splatting (3DGS).

Education

Ph.D. in Computer Science (Candidate) *2020 – 2026/2027 (Expected)*
University of Ghana, Legon Accra, Ghana

- *Dissertation:* **Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations**.
- *Supervisors:* Prof. Ebenezer Owusu (Principal Supervisor) and Prof. Justice Kwame Appati (Co-Supervisor).
- *Focus:* Epistemic reliability calibration, selective top-k semantic retention, 2D-to-3D geometric consensus, and process-supervised RLVR.

Master of Philosophy (MPhil) in Computer Science *2016*
University of Ghana, Legon Accra, Ghana

- *Thesis:* **Sentiment Analysis of Twitter Feeds using Machine Learning: Effect of Feature Hash-Bit Size**.

B.Sc. in Computer Science (GIMPA, 2012) **HND in Electrical & Electronic Engineering** (Accra Technical University, 2009)

Publications & Manuscripts Under Review

1. **Gah, S. K.**, Owusu, E., Appati, J. K. (2026). **Representation Before Weighting: Semantic Retention in Frozen 3D Foundation-Model Fusion**. *IEEE/CVF Computer Vision and Pattern Recognition (CVPR 2027)*, Paper #9692 [Under Review].
2. **Gah, S. K.**, Owusu, E., Appati, J. K., et al. (2026). **Reliable Cross-Modal Correction of Frozen Foundation Models for Training-Free Open-Vocabulary 3D Segmentation**. *Image and Vision Computing (IVC)*, Elsevier [Under Review].
3. **Gah, S. K.**, Owusu-Oware, E., Owusu, E., Appati, J. K. (2026). **Training-Free Generalized Few-Shot Segmentation with Frozen SAM3 and CLIP: How Much Does the Support Set Add?** *Machine Vision and Applications (MVA)*, Springer Nature [Under Review].
4. **Gah, S. K.**, Owusu-Oware, E., Owusu, E. (2026). **Training-Free Open-Vocabulary Point-Cloud Segmentation on Generalized Few-Shot Benchmarks**. *Asian Conference on Computer Vision (ACCV 2026)*, Paper #482.

Research Experience

Lead Doctoral Researcher — AI Systems & Visual Perception *2020 – Present*
Department of Computer Science, University of Ghana Accra, Ghana

- **Representation Before Weighting on ScanNet200:** Demonstrated that preserving compact Top-20 hypotheses recovers +7.02 HM over greedy argmax collapse on 156 ScanNet200 scenes (reaching 35.49 HM), whereas full-distribution retention incurs long-tail noise degradation (34.87 HM).
- **Multi-View Geometric Consensus on ScanNet++:** Developed cross-view geometric lifting pipelines projecting promptable 2D SAM3 masks into 3D coordinate space; eliminated occlusion overwrite errors, raising ScanNet++ novel-class mIoU from 16.16 to 32.20 (21.50 → 32.00 HM).
- **High-Throughput Sparse Serialization (2.08M pts/sec):** Engineered Coordinate (COO) sparse tensor serialization pipelines streaming 21.4 GiB of representation data on CPU memory budgets (RSS 719.5 MiB), enabling large-scale evaluation without enterprise GPU clusters.
- **Support-Set Benchmark Audit:** Audited generalized few-shot foundation model conditioning on SAM3 and CLIP, identifying that linguistic priors dominate visual support and resolving an aspect-ratio letterboxing confound of up to 39.96 HM points.

Founder & Principal Systems Architect *2020 – Present*
EED Soft Consult · Applied AI Research & High-Throughput Systems Accra, Ghana

- **Autonomous Multi-Agent Systems & MCP Tooling:** Architected multi-agent reasoning workflows using LangGraph and custom Model Context Protocol (MCP) servers, integrating AST parsers and verification guardrails for automated code analysis.
- **Production GenAI Legal Platform (GhanaLexAI):** Deployed an open-access platform indexing 10,000+ legal statutes and court rulings (ghanalexai.com); achieved sub-second latency via hybrid vector search with deterministic citation verification gates.

Teaching & Academic Mentorship

Course Lead & Lecturer — Systems Administration (Stanford SRE Standard) *2022 – Present*
Accra Technical University (ATU) · Department of Computer Science Accra, Ghana

- **Curriculum Architecture (BCP 307):** Designed and delivered a rigorous 10-week curriculum modeled after Stanford CS110 and Google Site Reliability Engineering standards, covering cloud VPC networking, containerization (Docker/Podman), Linux internals, and disaster recovery.
- **Observability & Telemetry Practicums:** Directed laboratory workshops training 150+ undergraduate engineering students in instrumenting Prometheus Node Exporters, centralized 'rsyslog' logging, and Grafana alert pipelines.
- **Underrepresented Scholar Mentorship:** Mentored 15+ undergraduate students through technical capstone projects, research dissertations, and industry certifications, with multiple mentees securing competitive software engineering roles.

Graduate Teaching Assistant *2014 – 2016*
Department of Computer Science, University of Ghana Accra, Ghana

- Assisted in teaching undergraduate courses in Data Structures, Algorithms, and Object-Oriented Programming (Java/C++).
- Supervised bi-weekly laboratory sessions, graded problem sets, and held weekly office hours for cohorts of 80+ students.

Public-Interest Systems Engineering Experience

Lead Enterprise Solutions Architect *2023 – Present*
National Ambulance Service (NAS) Digital Operations Platform Accra, Ghana

- **High-Availability Mission-Critical Architecture:** Spearheaded architecture and production delivery of a nationwide 24/7 digital emergency dispatch platform supporting 34 regional centers, maintaining **99.95% uptime** under stringent operational SLAs.
- **High-Throughput Microservices:** Designed 20+ secure RESTful API microservices (FastAPI / Node.js) with Redis in-memory caching and PostgreSQL clustering, processing real-time fleet telemetry with sub-100ms response times.

Technical Skills & Scientific Stack

Vision & AI Frameworks	PyTorch, TorchVision, Hugging Face, OpenScene, Segment Anything (SAM/SAM3), RegionPLC, DINOv2, CLIP, LangGraph, DeepEval, MLflow, Scikit-learn.
3D & Spatial Computing	Point Cloud Processing (Open3D, PyTorch3D), ScanNet200, ScanNet++, Coordinate (COO) Sparse Tensors, Multi-View Camera Projection, Epipolar Geometry.
Languages	Python (Expert, AsyncIO, NumPy, SciPy), Modern C++17, TypeScript / Node.js, SQL (PostgreSQL), Bash/Linux Shell.
Cloud, DevOps & SRE	Docker, Kubernetes, Podman, AWS (Certified Cloud Practitioner), Prometheus, Grafana, GitHub Actions CI/CD, Linux Kernel Internals.

Awards, Honors & Professional Credentials

- **AWS Certified Cloud Practitioner** — Amazon Web Services (Validation of enterprise cloud architecture).
- **Stanford SRE Practicum Standard** — Linux Internals, Cloud VPC, Distributed Systems Telemetry.
- **PhD Research Seminar Speaker** — Methodist University Ghana (Nov 2026).
- **Postgraduate Research Grant Recipient** — University of Ghana Doctoral Research Fund.

Academic References

Available upon request, including references from the Department of Computer Science (University of Ghana), Colorado State University, and institutional collaborators.